Clinical interpretation — sub-M2030 ses-1454

Chronic (>6 mo) 3839 days post-onset

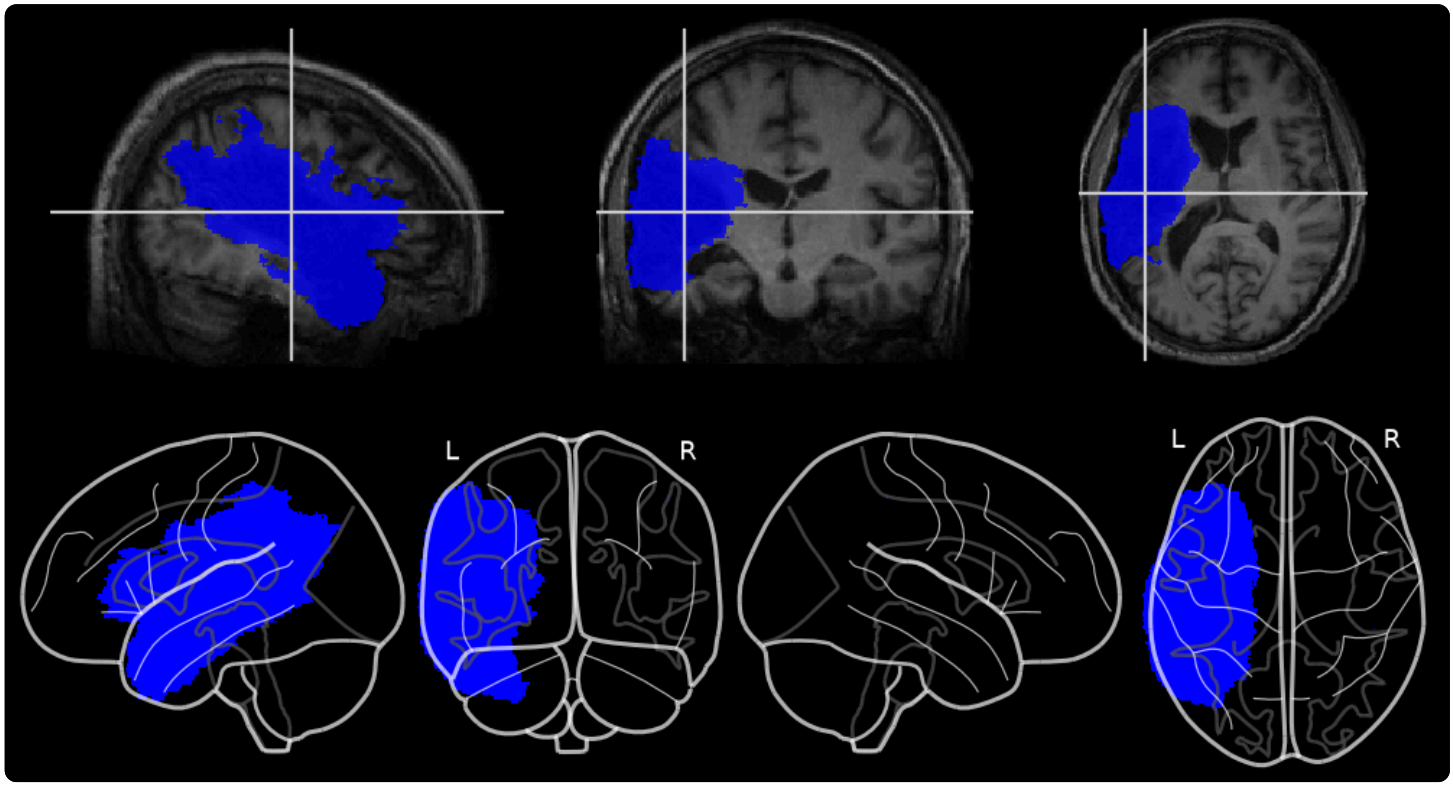
Lesion mask: **Manual** · research use only, not a diagnosis
KB scope: approved, in_review, draft · (all confidence levels included) · atlas (lookup): AAL, HarvardOxford, Destrieux, Schaefer400, JHU, XTRACT · atlas (display): HarvardOxford

QC status: No QC checkpoint choice is recorded.

Research use only — not a diagnosis. This report combines automated lesion segmentation with literature associations from the aphasia knowledge base. It does not diagnose aphasia, predict an individual's outcome or prescribe therapy; any clinical use requires review by a qualified clinician.

- **Lesion mask:** Manual (MNI) `ExpertMask_in_MNI.nii.gz`. QC: not rated in the QC checkpoint.
- **How this lesion mask was made:**
  1. Expert-drawn mask sub-M2030_ses-1455_acq-spc3_run-3_T2w_desc-lesion_mask.nii.gz, drawn on sub-M2030_ses-1455_acq-spc3_run-3_T2w.nii.gz; that scan was rigidly registered to the T1w sub-M2030_ses-1454_acq-tfl3p2_run-3_T1w.nii.gz (SimpleITK, Mattes mutual information, seed 1729) and the mask resampled with nearest neighbour.
  2. Warped to the FSL MNI152 template (MNI152NLin6Asym, 1 mm) with this T1w's ANTs registration (ants/2.6.5: affine, then SyN, with the SynthStroke lesion excluded from the cost function, seed 1729), nearest-neighbour resampling.
- **Template space:** Lesion masks are on the FSL MNI152 template (MNI152NLin6Asym, 1 mm); the registration used for this mask is named above. Atlases are resampled to the lesion by world coordinates. Overlaps still depend on how well this individual brain was registered to MNI152, which matters most for small or thin regions.
- **Knowledge base:** version b9990f5 (2026-10-21) · 97 papers · 461 extracted findings (47 approved by a named reviewer) in 246 entries (158 regions, 18 impairments, 19 therapies, 51 predictors) · statuses used: approved, in_review, draft
- **Generated:** 2026-10-21 20:39 · KB scope: approved, in_review, draft (all confidence levels included)

Lesion viewer — sub-M2030 ses-1454 (MNI space) ■ Manual (blue) Manual lesion



Damaged regions (HarvardOxford, ≥5.0%)

HarvardOxford region	Lesion side	% in ROI	
Temporal Pole	Left	7.4%	1,833 vox

Lesion side is measured from the actual mask overlap; bilateral entries show left/right counts.

Behavioural / clinical data

WAB-AQ	51.5
WAB type	Broca [flu − · comp + · rep −]
Age at stroke	54
Sex	M
Days to WAB	3839
Lesion volume	199.2 mL

Mapped to KB predictors: severity_metric, age, time_post_onset, sex, lesion_volume

Lesion location: what the evidence says

A region counts as damaged when it holds ≥5% of the lesion or ≥25% of it is destroyed. Regions are matched to the knowledge base through the curated atlas mapping (`aphasia-kb/atlas_mapping/`). Of the knowledge base's 318 location findings, **234** concern regions this lesion damages, 67 concern regions it does not damage, and 17 concern regions no atlas can represent (e.g. the extreme capsule).

Damage to these regions was associated with	Findings	Evidence	Damaged regions involved	Citations
Semantics (semantic processing) [impairment]	29 associations · 2 no association	□□□□ very strong / replicated	Left Angular Gyrus, Left Anterior Temporal Lobe, Left IFG Pars Triangularis (BA 45), Left Inferior Fronto-Occipital Fasciculus, Left Inferior Longitudinal Fasciculus, Left Middle Longitudinal Fasciculus, Left Middle Temporal Gyrus (Ventral Stream Node), Left Parietal Association Cortex, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale), Left Superior Temporal Sulcus, Left Uncinate Fasciculus, Left anterior middle temporal gyrus, Left temporal pole	@Alyahya2018NounVerb, @Alyahya2020, @Bates2003, @Binder2009, @Breier2008, @Butler2014, @Catani2008, @DeWitt2012, @Dick2012, @Dronkers2004, @Fridriksson2013, @Halai2017, @Ivanova2016, @Schwartz2009
Phonological production [impairment]	23 associations · 1 no association	□□□□ very strong / replicated	Left Angular Gyrus, Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Inferior Longitudinal Fasciculus, Left Posterior Superior Temporal Gyrus, Left Superior Longitudinal Fasciculus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Supramarginal Gyrus, Left Supramarginal Gyrus and Premotor Cortex (supra-Sylvian), Left arcuate fasciculus, Left parietal operculum cortex, Left posterior supramarginal gyrus	@Alyahya2018NounVerb, @Alyahya2020, @Breier2008, @Butler2014, @Dick2012, @Halai2017, @Henseler2014, @Ivanova2016, @Kummerer2013, @Schwartz2012
Broca's aphasia [impairment]	15 associations	□□□□ very strong / replicated	Left Heschl's Gyrus, Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Superior Temporal Gyrus (Dorsal Stream Node), Left Supramarginal Gyrus, Left anterior arcuate fasciculus, Left arcuate fasciculus, Left rolandic operculum	@Henseler2014, @Yourganov2015Predicting
Form-to-meaning processing (ventral stream) [impairment]	15 associations	□□□□ very strong / replicated	Left Angular Gyrus, Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Uncinate Fasciculus, Left anterior middle temporal gyrus, Left arcuate fasciculus, Left posterior middle temporal gyrus, Left posterior supramarginal gyrus	@Breier2008, @Fridriksson2013, @Fridriksson2018, @Henseler2014
Phonological recognition [impairment]	9 associations	□□□□ very strong / replicated	Left Heschl's Gyrus, Left Posterior Planum Temporale (Area Spt), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node)	@Alyahya2018NounVerb, @Alyahya2020, @DeWitt2012, @Thye2018
Fluency (speech fluency) [impairment]	14 associations · 3 no association	□□□□ very strong / replicated	Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left IFG Pars Triangularis (BA 45), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale), Left Uncinate Fasciculus, Left anterior arcuate fasciculus, Left arcuate fasciculus, Left precentral gyrus of the insula, Left rolandic operculum	@Alyahya2018NounVerb, @Bates2003, @Bonilha2014, @Fridriksson2013, @GajardoVidal2021, @Hope2013, @Yourganov2016
Global aphasia [impairment]	12 associations	□□□□ very strong / replicated	Left IFG Pars Triangularis (BA 45), Left Uncinate Fasciculus, Left anterior arcuate fasciculus, Left precentral gyrus of the insula, Left putamen, Left temporal pole MTG	@Yourganov2015Predicting
Syntactic Processing Deficits in	6 associations · 3 no	□□□□ very strong / replicated	Left Angular Gyrus, Left Anterior Temporal Lobe, Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left IFG Pars Triangularis	@Dronkers2004, @Matchin2020, @Rogalsky2018

Aphasia [impairment]	association, 1 mixed results		(BA 45), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale)	
Speech repetition [impairment]	8 associations	□□□□ very strong / replicated	Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway), Left IFG Pars Opercularis (Broca's area, dorsal stream hub), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Dorsal Stream Node), Left Supramarginal Gyrus, Left Temporoparietal Junction, Left arcuate fasciculus	@Fridriksson2010, @Saur2008, @Yourganov2016
Auditory comprehension [impairment]	6 associations	□□□□ very strong / replicated	Left Middle Temporal Gyrus (Ventral Stream Node), Left Posterior Superior Temporal Gyrus, Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale), Left Superior Temporal Sulcus	@Bonilha2014, @DeWitt2012, @Robson2019, @Saur2008, @Yourganov2016

Evidence by damaged region (click a region to see each finding; □ marks regions whose evidence comes from 2 studies or fewer):

- ▶ **Left Posterior Superior Temporal Gyrus** (left) · 3.5% of the lesion · 100% of the region destroyed · 20 findings
- ▶ **Left Posterior Planum Temporale (Area Spt)** (left) □ □ based on 2 studies · 2.1% of the lesion · 100% of the region destroyed · 4 findings
- ▶ **Left parietal operculum cortex** (left) □ based on 1 study · 2.3% of the lesion · 100% of the region destroyed · 2 findings
- ▶ **Left Heschl's Gyrus** (left) · 1.3% of the lesion · 100% of the region destroyed · 7 findings
- ▶ **Left rolandic operculum** (left) □ based on 2 studies · 4.0% of the lesion · 100% of the region destroyed · 3 findings
- ▶ **Left Superior Temporal Gyrus (Wernicke's Area and Planum Temporale)** (left) □ · 5.7% of the lesion · 100% of the region destroyed · 13 findings
- ▶ **Left Superior Temporal Gyrus (Dorsal Stream Node)** (left) · 4.5% of the lesion · 100% of the region destroyed · 12 findings
- ▶ **Left anterior middle temporal gyrus** (left) □ based on 2 studies · 1.8% of the lesion · 99% of the region destroyed · 3 findings
- ▶ **Left anterior supramarginal gyrus** (left) □ based on 1 study · 3.7% of the lesion · 97% of the region destroyed · 2 findings
- ▶ **Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway)** (left) · 1.1% of the lesion · 96% of the region destroyed · 6 findings
- ▶ **Left Superior Longitudinal Fasciculus** (left) □ based on 1 study · 1.1% of the lesion · 96% of the region destroyed · 1 finding
- ▶ **Left precentral gyrus of the insula** (left) □ · 4.5% of the lesion · 96% of the region destroyed · 10 findings
- ▶ **Left arcuate fasciculus** (left) □ · 5.0% of the lesion · 90% of the region destroyed · 26 findings
- ▶ **Left Supramarginal Gyrus** (left) · 7.1% of the lesion · 88% of the region destroyed · 5 findings
- ▶ **Left Temporoparietal Junction** (left) □ □ based on 2 studies · 9.8% of the lesion · 85% of the region destroyed · 3 findings
- ▶ **Left anterior arcuate fasciculus** (left) □ · 7.3% of the lesion · 84% of the region destroyed · 7 findings
- ▶ **Left Anterior Temporal Lobe** (left) □ · 10.1% of the lesion · 83% of the region destroyed · 8 findings
- ▶ **Left posterior supramarginal gyrus** (left) □ based on 2 studies · 3.4% of the lesion · 80% of the region destroyed · 4 findings
- ▶ **Left Superior Temporal Sulcus** (left) □ □ based on 2 studies · 6.8% of the lesion · 79% of the region destroyed · 2 findings
- ▶ **Left temporal pole** (left) □ based on 1 study · 7.4% of the lesion · 78% of the region destroyed · 1 finding
- ▶ **Left Angular Gyrus** (left) · 2.9% of the lesion · 77% of the region destroyed · 14 findings
- ▶ **Left temporal pole MTG** (left) □ based on 1 study · 2.3% of the lesion · 76% of the region destroyed · 3 findings
- ▶ **Left Anterior Temporal Lobe (Residual Comprehension Substrate in Wernicke's Aphasia)** (left) □ □ based on 1 study · 10.2% of the lesion · 75% of the region destroyed · 4 findings
- ▶ **Left Middle Longitudinal Fasciculus** (left) □ based on 1 study · 3.9% of the lesion · 72% of the region destroyed · 1 finding
- ▶ **Left temporal pole STG** (left) □ based on 1 study · 3.7% of the lesion · 71% of the region destroyed · 1 finding
- ▶ **Left Temporal Lobe** (left) □ based on 1 study · 23.9% of the lesion · 67% of the region destroyed · 1 finding
- ▶ **Left Middle Temporal Gyrus (Ventral Stream Node)** (left) · 6.8% of the lesion · 66% of the region destroyed · 9 findings
- ▶ **Left posterior middle temporal gyrus** (left) □ based on 1 study · 3.3% of the lesion · 64% of the region destroyed · 2 findings
- ▶ **Left putamen** (left) □ based on 1 study · 2.5% of the lesion · 62% of the region destroyed · 3 findings

▶ **Left IFG Pars Opercularis (Broca's area, dorsal stream hub)** (left) · 1.8% of the lesion · 60% of the region destroyed · 22 findings

▶ **Left Parietal Association Cortex** (left) □ □ based on 1 study · 10.2% of the lesion · 58% of the region destroyed · 1 finding

▶ **Left Uncinate Fasciculus** (left) · 2.6% of the lesion · 57% of the region destroyed · 9 findings

▶ **Left Inferior Fronto-Occipital Fasciculus** (left) · 0.1% of the lesion · 56% of the region destroyed · 3 findings

▶ **Left IFG Pars Triangularis (BA 45)** (left) · 1.3% of the lesion · 52% of the region destroyed · 11 findings

▶ **Left Supramarginal Gyrus and Premotor Cortex (supra-Sylvian)** (left) □ · 11.2% of the lesion · 50% of the region destroyed · 4 findings

▶ **Left Inferior Longitudinal Fasciculus** (left) · 0.1% of the lesion · 47% of the region destroyed · 6 findings

▶ **Left Postcentral Gyrus** (left) □ based on 1 study · 4.1% of the lesion · 28% of the region destroyed · 1 finding

White-matter disconnection: what the evidence says

The BCBToolkit map gives, for every voxel, the probability that its fibres pass through the lesion. A tract counts as disconnected when ≥50% of its voxels have a probability above 0.5. This catches tracts cut by the lesion away from where it lies, which direct overlap misses.

▶ **Left Inferior Fronto-Occipital Fasciculus** (left) · 100% of the tract disconnected · also directly damaged (56% destroyed) · 3 findings

▶ **Left Arcuate Fasciculus and Superior Longitudinal Fascicle (Dorsal Language Pathway)** (left) · 100% of the tract disconnected · also directly damaged (96% destroyed) · 6 findings

▶ **Left Superior Longitudinal Fasciculus** (left) □ based on 1 study · 100% of the tract disconnected · also directly damaged (96% destroyed) · 1 finding

▶ **Left Inferior Longitudinal Fasciculus** (left) · 90% of the tract disconnected · also directly damaged (47% destroyed) · 6 findings

▶ **Left arcuate fasciculus** (left) · 82% of the tract disconnected · also directly damaged (90% destroyed) · 26 findings

▶ **Left optic radiations** (left) □ based on 1 study · 69% of the tract disconnected · also directly damaged (7% destroyed) · 1 finding

▶ **Left Middle Longitudinal Fasciculus** (left) □ based on 1 study · 58% of the tract disconnected · also directly damaged (72% destroyed) · 1 finding

▶ **Left anterior arcuate fasciculus** (left) · 52% of the tract disconnected · also directly damaged (84% destroyed) · 7 findings

This participant's measures: what the evidence says

Findings about measures such as lesion volume or aphasia severity report associations (e.g. larger lesions with more severe aphasia), not predictions for one person.

Measure (this participant)	What the evidence says
Aphasia severity WAB-AQ 51.5 (moderate, Kertesz bands), measured 3839 days after the stroke	Lower values go with worse language or recovery. (13 findings: 13 association) ▶ Show the findings □ 9 of these findings concern severity measured early after the stroke (or recovery from it); this score was measured 3839 days after the stroke, so they apply only loosely.
Age at stroke 54 years	Higher values go with worse language or recovery. (4 findings: 2 association, 1 inverse association, 1 no association) ▶ Show the findings
Time since stroke 3839 days	Direction: see each finding. (4 findings: 3 association, 1 no association) ▶ Show the findings
Sex M	Direction: see each finding. (3 findings: 2 association, 1 no association) ▶ Show the findings
Lesion volume 199 ml (this report's lesion mask)	Higher values go with worse language or recovery. (19 findings: 14 association, 4 no association, 1 inverse association) ▶ Show the findings

Therapy recommendations (5 matched)

Therapies with evidence targeting the predicted impairments above.

Therapy	Addresses impairment(s)	Active ingredients (RTSS)	Brief description	Citations
Constraint-Induced Aphasia Therapy (CIAT / CILT)	**Broca's aphasia** (behavioural); Anomic aphasia, Broca's aphasia, Fluency (speech fluency), Form-to-meaning processing (ventral stream), Global aphasia (lesion)	Verbal Constraint Communicative Shaping Massed Practice	An intensive, massed-practice therapy in which compensatory strategies (gesture, drawing, writing) are prohibited and th…	@Maher2006, @Pulvermuller2001
Melodic Intonation Therapy (MIT)	**Broca's aphasia** (behavioural); Broca's aphasia, Fluency (speech fluency), Global aphasia, Phonological production (lesion)	Melodic Intoning Rhythmic Left-Hand Tapping Communicative Shaping Massed Practice	A therapy that uses melodic intoning (singing on two pitches) and rhythmic left-hand tapping to elicit propositional spe…	@Schlaug2008, @Stahl2013, @vanderMeulen201 @vanderMeulen201
Melodic Intonation Therapy	**Broca's aphasia** (behavioural); Fluency (speech fluency), Phonological production (lesion)	—	A language production treatment for severe nonfluent aphasia involving repetitive singing of short sentences combined wi…	@Schlaug2008, @Stahl2013, @vanderMeulen201 @vanderMeulen201
Treatment of Underlying Forms (TUF)	**Broca's aphasia** (behavioural); Broca's aphasia, Syntactic Processing Deficits in Aphasia (lesion)	Verb Argument Structure Training	A linguistically-motivated, impairment-based intervention for agrammatic aphasia that uses meta-linguistic steps focused…	@Barbieri2023, @Thompson2003
Inhibitory rTMS to Right Pars Triangularis (BA45) for Anomia in Nonfluent Aphasia	**Broca's aphasia** (behavioural); Anomic aphasia, Broca's aphasia, Fluency (speech fluency) (lesion)	Inhibitory rTMS (Right Pars Triangularis / BA45)	Open-protocol pilot study applying slow (1 Hz) inhibitory repetitive TMS daily for 10 days to the right pars triangulari…	@Naeser2005

procedure dose parameter technology

Legend.

Evidence strength: □ weak · □□ moderate · □□□ strong · □□□□ very strong. Hover stars for raw score (bins are a project display heuristic, not an evidence grading).

Finding direction: what the study found between the region or measure and the outcome: association, inverse association, no association or mixed results. A finding reports an association in a group of patients, not a prediction for this person. □ = a region matched to the atlas by a judgement call (`aphasia-kb/atlas_mapping/README.md`).

WAB decomposition: flu = fluency, comp = comprehension, rep = repetition. + preserved, − impaired (WAB type taxonomy).

Stage badge: Acute <2 wk · Subacute 2 wk–6 mo · Chronic >6 mo □ = therapy evidence from a different recovery stage (hover for detail).

Rules and their sources: `aphasia-kb/aphasia_rules.py`.